

Global Weather Analysis & Forecasting

Assessment

Harshitha Gaddambachahalli Raghukumar

1 Introduction

Weather forecasting plays a critical role in agriculture, transportation, disaster preparedness, aviation, environmental planning and day-to-day decision making. Accurate forecasting models help governments, businesses and individuals respond effectively to changing atmospheric conditions while reducing uncertainty and operational risk.

The objective of this project is to analyze a large-scale global weather dataset and develop machine learning models capable of forecasting future temperature trends. In addition to predictive modeling, the project explores climate patterns, air quality relationships, geographical weather variations and anomaly detection to generate meaningful insights from real world weather data.

This assessment was completed as part of the PM Accelerator internship evaluation process. PM Accelerator's mission is to empower future product, AI and technology leaders through hands on learning experiences, real-world projects and career acceleration opportunities that help candidates build impactful solutions and grow professionally.

The project follows a complete data science workflow consisting of data cleaning, exploratory data analysis, feature engineering, baseline time series forecasting, advanced model benchmarking, ensemble forecasting and domain specific analytical studies. The final outcome demonstrates both technical forecasting capability and business oriented analytical thinking.

2 Dataset Description

The dataset used in this project is the **Global Weather Repository**, sourced from Kaggle. It contains weather observations collected from cities around the world, providing a large and geographically diverse dataset suited for climate analysis and forecasting tasks. Observations span multiple countries and geographic locations, capturing a wide range of atmospheric conditions.

Key variables available in the dataset include temperature ($^{\circ}\text{C}$ and $^{\circ}\text{F}$), humidity, wind speed (kph), atmospheric pressure (mb), precipitation (mm), air quality indicators (PM2.5, PM10, Carbon Monoxide, Ozone, Nitrogen Dioxide), latitude, longitude and country-level location information. The `last_updated` timestamp column enables time-based

trend analysis and time series forecasting.

This dataset is suited for machine learning because it combines temporal, environmental and geographical information in a single source. The variety of features allows both predictive modeling and broader analytical tasks such as anomaly detection, climate comparison and regional weather pattern analysis.

3 Basic Assessment

3.1 Data Cleaning and Preprocessing

Real-world weather datasets often contain inconsistencies such as outliers, corrupt records and mixed data formats. An extensive preprocessing pipeline was applied to improve data quality before performing analysis and model development.

Domain-based filtering was used to remove physically impossible weather conditions. Observations where temperature exceeded the range of -50°C to 60°C , wind speed exceeded 200 kph, atmospheric pressure fell outside 870 - 100 mb or humidity fell outside the 0–100% range were removed. After applying this filtering, only 7 rows were removed from the dataset, confirming that the original data was already of high quality with very few invalid or corrupt records.

The `last_updated` column was converted into datetime format to support time series analysis and date-based feature engineering. Lag features (lag-1 and lag-7 temperature values) and a 7-day rolling mean were subsequently derived from the time-ordered series to provide temporal context for machine learning models.

3.2 Exploratory Data Analysis (EDA)

A total of six exploratory visualizations were created to analyze distributions, temporal trends, correlations and geographical weather patterns. The three most informative visualizations are discussed in detail below.

1. Temperature Distribution

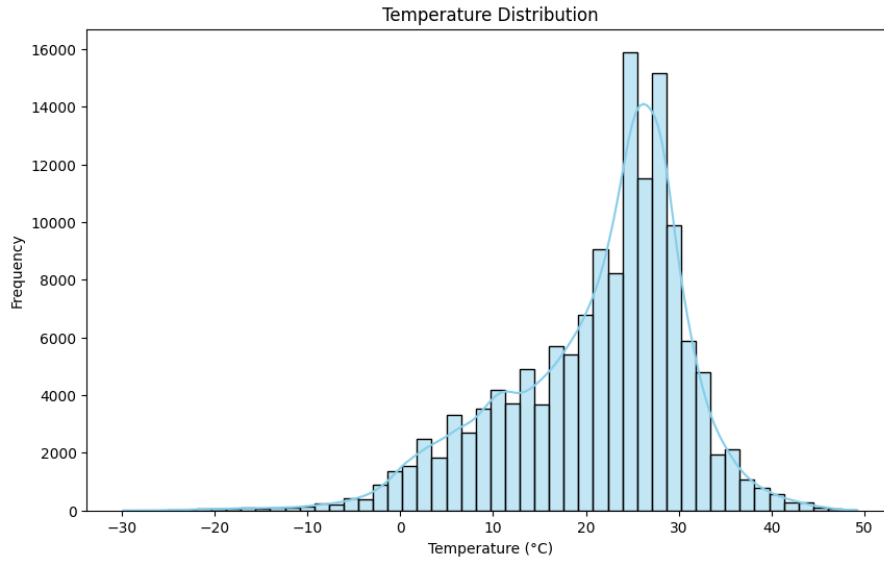


Figure 1: Temperature Distribution

The temperature distribution histogram reveals that global observations are broadly spread across a moderate range of values, reflecting the natural climatic diversity across countries and seasons. A KDE curve overlaid on the histogram shows the overall shape of the distribution, which is approximately bell-shaped but with slight skewness due to the presence of both arid and polar regions in the dataset.

2. Precipitation Distribution

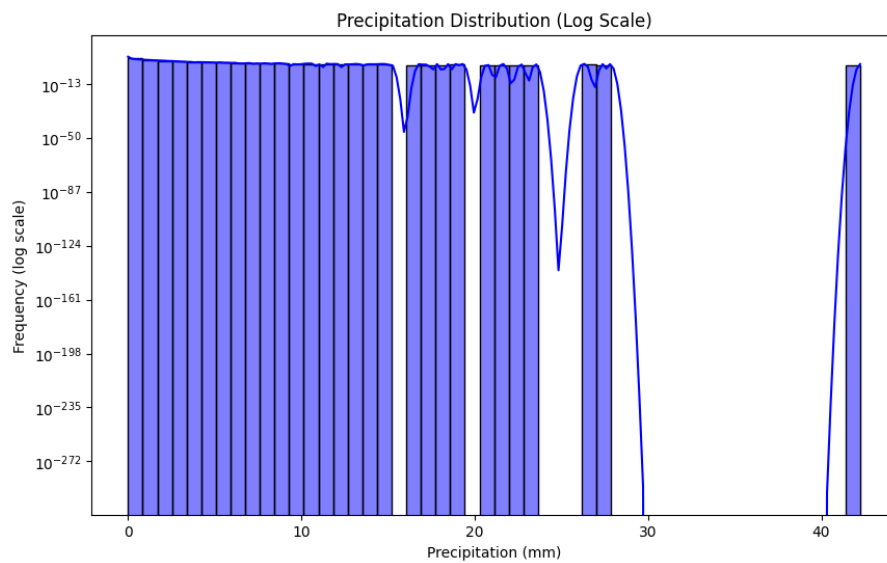


Figure 2: Precipitation Distribution (Log Scale)

Precipitation data is highly right-skewed, with most observations recording zero or near-zero rainfall. A logarithmic scale was applied to the y-axis to better visualize the full distribution. This skewness is expected given that a large proportion of global weather recordings come from dry or semi-arid regions where rainfall is infrequent.

3. Feature Correlation Heatmap

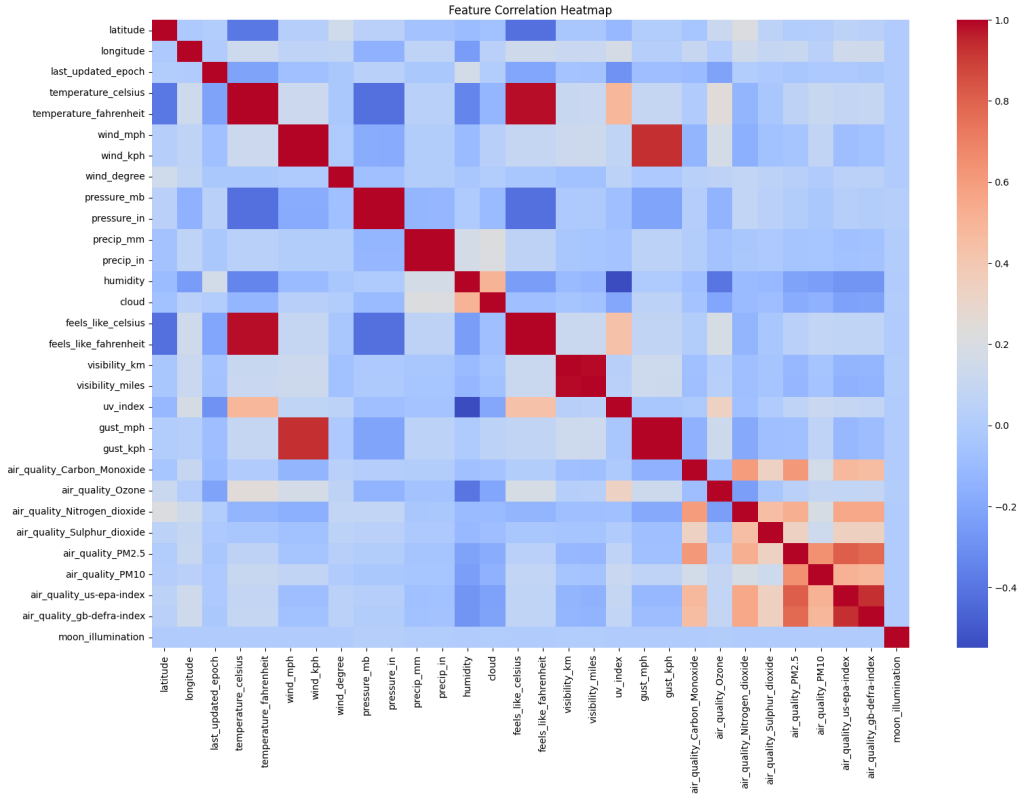


Figure 3: Feature Correlation Heatmap

The correlation heatmap was used to study relationships among all numerical weather variables. Strong positive correlations were observed between temperature and feels-like temperature, as well as between wind speed in kph and mph (unit equivalents). Air quality indicators such as PM2.5 and PM10 also showed strong mutual correlations. These findings helped identify multicollinear feature groups and guided feature selection for downstream modeling.

Additional visualizations including the violin plot of temperature, scatter plot of temperature versus precipitation and time series plots of temperature and precipitation trends over time were also generated to support the broader exploratory analysis.

3.3 Baseline Time Series Forecasting

To establish an initial forecasting benchmark, three simple baseline models were developed using daily average temperature aggregated from the `last_updated` timestamp. The dataset was divided chronologically into training and testing sets, with 80% of the observations used for training and the remaining 20% reserved for testing. This approach preserved the natural temporal order of the data and avoided future information leakage.

The three baseline models evaluated were:

- **Naive (Last Value):** Repeats the final training observation for all future time steps.
- **Seasonal Naive:** Repeats the last 7-day window from the training set, cycling through the test period.
- **Rolling Mean:** Forecasts all test steps using the 7-day trailing average of the training series.

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). All three baseline models produced negative R^2 scores, indicating that simple persistence-based methods are insufficient for capturing the complex temporal patterns present in global weather data. Among the baselines, the Rolling Mean model achieved the lowest MAE and RMSE values, suggesting that short-term temporal smoothing provides a modest improvement over a single-point forecast.

	Model	MAE	RMSE	R2
0	Naive (Last Value)	6.633181	8.501035	-0.205579
1	Seasonal Naive	6.594846	8.384853	-0.172851
2	Rolling Mean	6.195595	7.944258	-0.052831

Figure 4: Baseline Temperature Forecasting

4 Advanced Assessment

4.1 Advanced EDA: Anomaly Detection

To identify unusual weather observations, an anomaly detection analysis was performed using three complementary methods: Z-Score, Interquartile Range (IQR) and Isolation Forest.

Z-Score flagged temperature observations that deviated more than three standard de-

viations from the mean, capturing extreme univariate outliers. **IQR-based detection** flagged observations outside the $1.5 \times \text{IQR}$ boundary computed from the first and third quartiles of temperature. These two methods focus purely on univariate statistical deviations and are effective for identifying isolated extreme values.

Isolation Forest was applied as a multivariate anomaly detector across five key features: temperature, humidity, wind speed, atmospheric pressure and precipitation, with a contamination rate of 1%. Unlike the univariate methods, Isolation Forest is capable of detecting anomalies that arise from unusual combinations of variables, making it more appropriate for the complexity of global weather data.

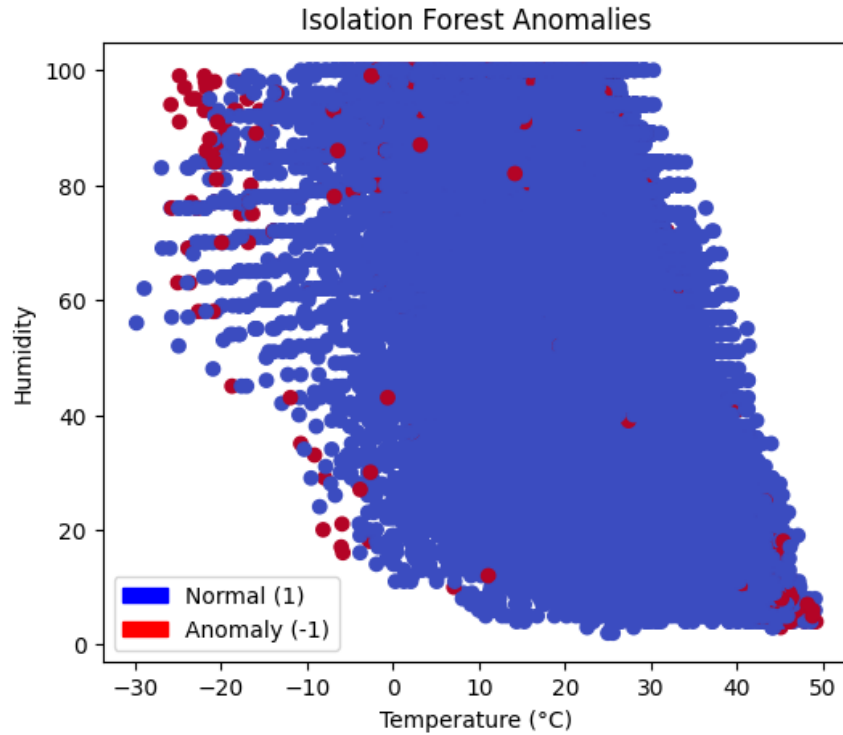


Figure 5: Isolation Forest Anomaly Detection Temperature vs. Humidity

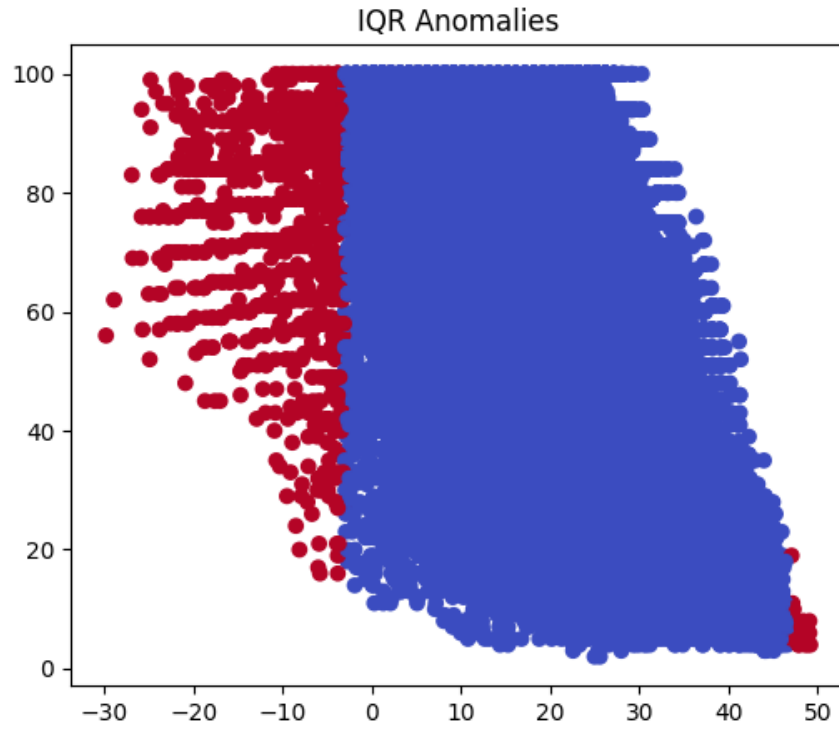


Figure 6: IQR Anomaly Detection Temperature vs. Humidity

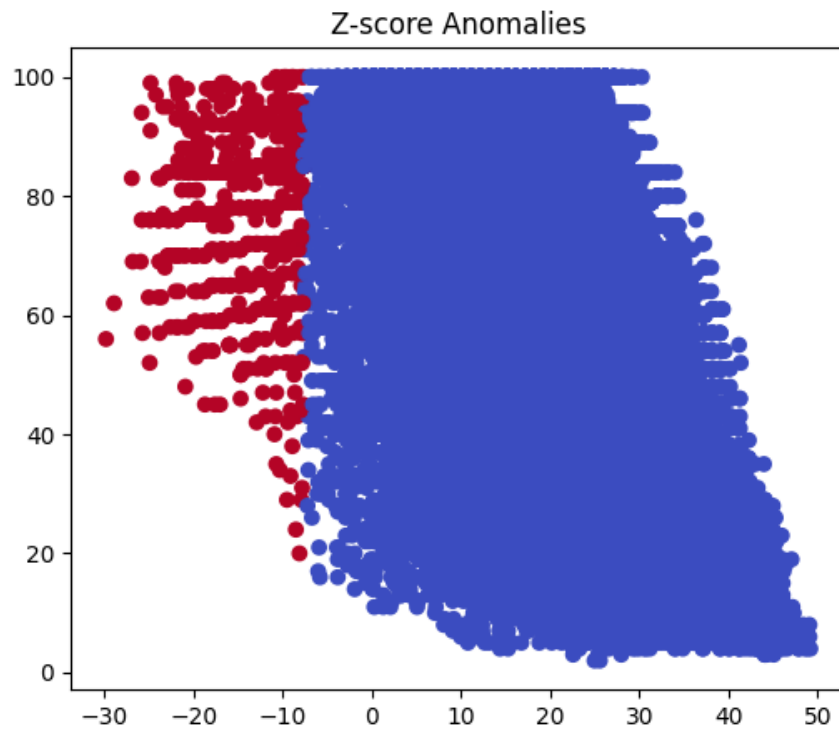


Figure 7: Z-Score Anomaly Detection Temperature vs. Humidity

The three methods highlight different subsets of unusual data points. IQR and Z-Score methods identify statistically extreme individual observations, while Isolation Forest

captures records that are anomalous across multiple dimensions simultaneously. This makes Isolation Forest the most robust approach for complex, high-dimensional weather datasets. The detected anomalies may indicate rare climate events, abrupt atmospheric changes or extreme environmental conditions.

4.2 Forecasting with Multiple Models

To improve forecasting performance beyond the baseline, several advanced time series and machine learning models were trained and evaluated. Lag features (lag-1, lag-7 temperature values and a 7-day rolling mean) were engineered from the daily temperature series and used as predictors for all machine learning models. The same 80/20 chronological train-test split was maintained throughout.

The models compared in this study were:

- **ARIMA(5,1,0):** A classical autoregressive integrated moving average model fit to the training series.
- **Prophet:** Facebook’s open-source time series model designed to capture seasonality and trend components.
- **Random Forest Regressor:** An ensemble of 100 decision trees trained on the engineered lag features.
- **Gradient Boosting Regressor:** A sequential boosting approach with 100 estimators, learning rate 0.1 and max depth 3.
- **XGBoost Regressor:** An optimized gradient boosting framework with 100 estimators, learning rate 0.1 and max depth 5.

Model performance was measured using MAE, RMSE and R^2 .

	Model	MAE	RMSE	R2
3	Gradient Boosting	4.780249	6.262854	0.345898
4	XGBoost	4.811903	6.295794	0.339000
2	Random Forest	5.061595	6.574183	0.279251
0	ARIMA	6.354954	8.157356	-0.110071
1	Prophet	116.671477	135.502895	-305.301569

Figure 8: Multiple Model Performance

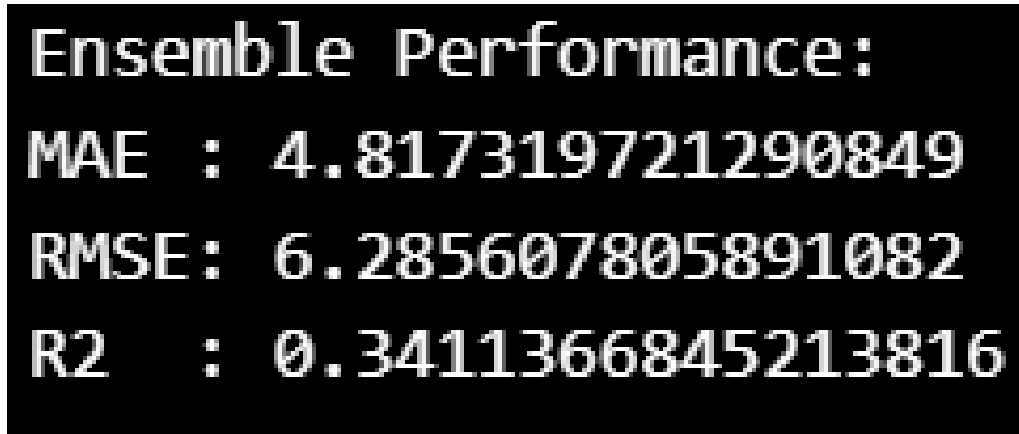


Figure 9: Ensemble Model

Among all models evaluated, tree-based machine learning methods substantially outperformed classical statistical approaches. **Gradient Boosting** achieved the best individual performance, with the lowest MAE and RMSE and the highest R^2 score, closely followed by XGBoost. Random Forest also performed competitively but was marginally less accurate than the boosting-based methods.

ARIMA and Prophet both produced negative R^2 values. ARIMA's poor performance reflects the violation of its stationarity and linearity assumptions by this complex, multi-region dataset. Prophet similarly struggled due to the irregular and globally aggregated nature of the series, which lacks the single consistent seasonal trend that Prophet is designed to model.

Ensemble Model

An ensemble forecast was constructed by averaging predictions from Random Forest, Gradient Boosting and XGBoost. This approach aimed to combine the strengths of the three best-performing individual models and improve overall forecasting robustness.

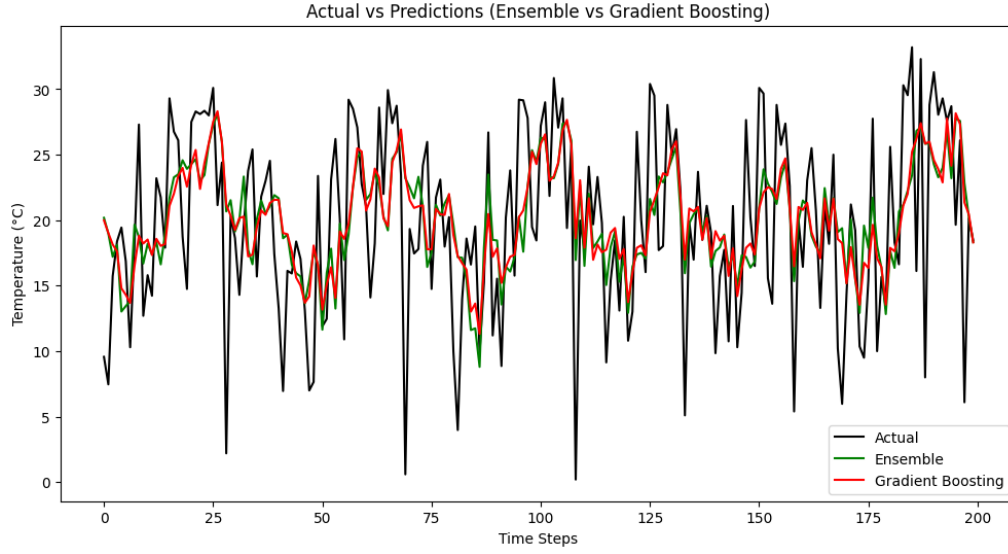


Figure 10: Actual vs. Predicted Temperature for Ensemble vs. Gradient Boosting

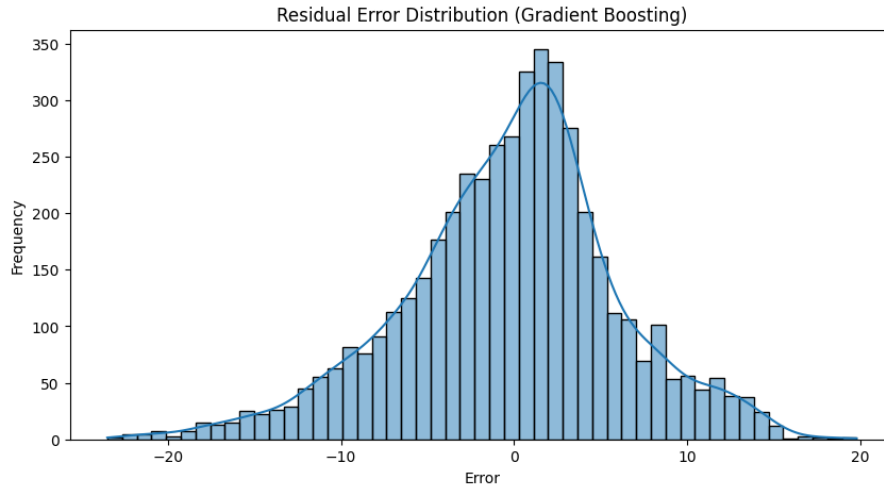


Figure 11: Residual Error Distribution Gradient Boosting

The Ensemble model delivered stable and balanced performance; however, it did not significantly outperform Gradient Boosting in isolation, suggesting that Gradient Boosting already captures the majority of the underlying predictive structure in the data. The residual error distribution for Gradient Boosting is approximately centred around zero, indicating that the model does not exhibit systematic bias.

5 Unique Analyses

5.1 Climate Analysis

To explore long-term climate behaviour, monthly global average temperature values were calculated and analyzed over time. This aggregation smooths short-term daily fluctua-

tions and reveals broader seasonal and temporal climate patterns present in the dataset.

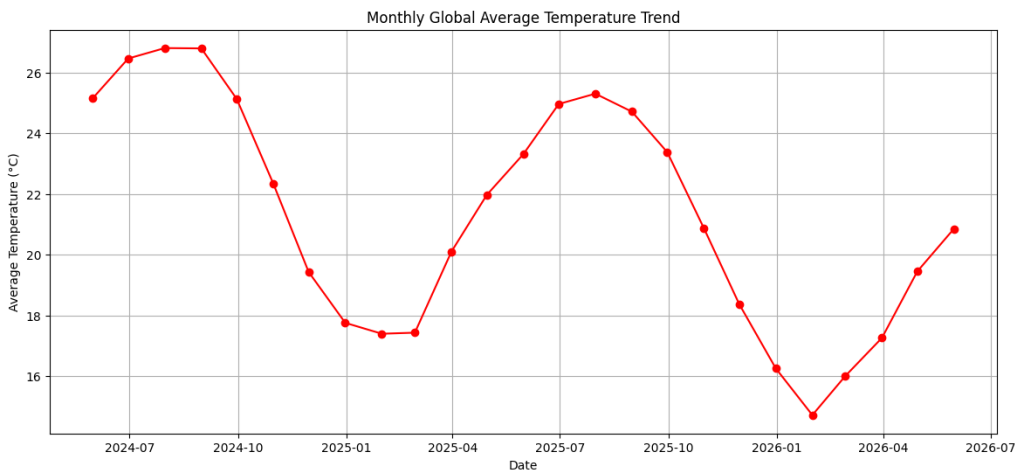


Figure 12: Monthly Global Average Temperature Trend

The monthly trend reveals cyclical temperature movement consistent with global seasonal patterns. Warmer periods are followed by cooler intervals, confirming expected climate seasonality and demonstrating that time-based forecasting is an appropriate strategy for weather prediction tasks.

Country-level climate comparisons were also performed to identify the warmest and rainiest regions in the dataset.

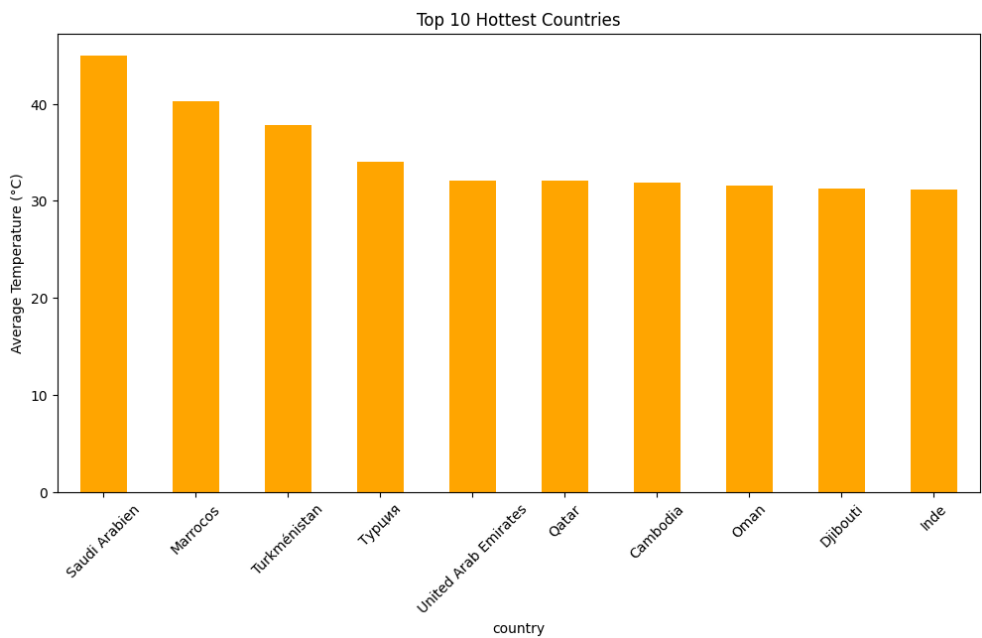


Figure 13: Top 10 Hottest Countries by Average Temperature

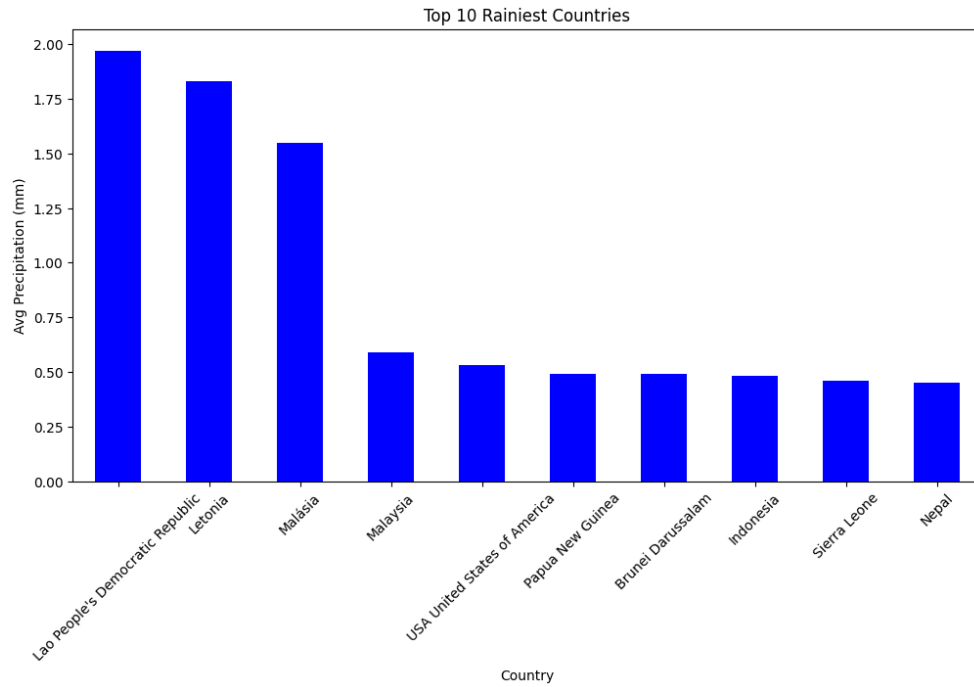


Figure 14: Top 10 Rainiest Countries by Average Precipitation

The results indicate that arid and desert-climate nations recorded the highest average temperatures, while tropical and island nations experienced the greatest rainfall levels. These findings align with well-established global climate geography and confirm the diversity captured within the dataset.

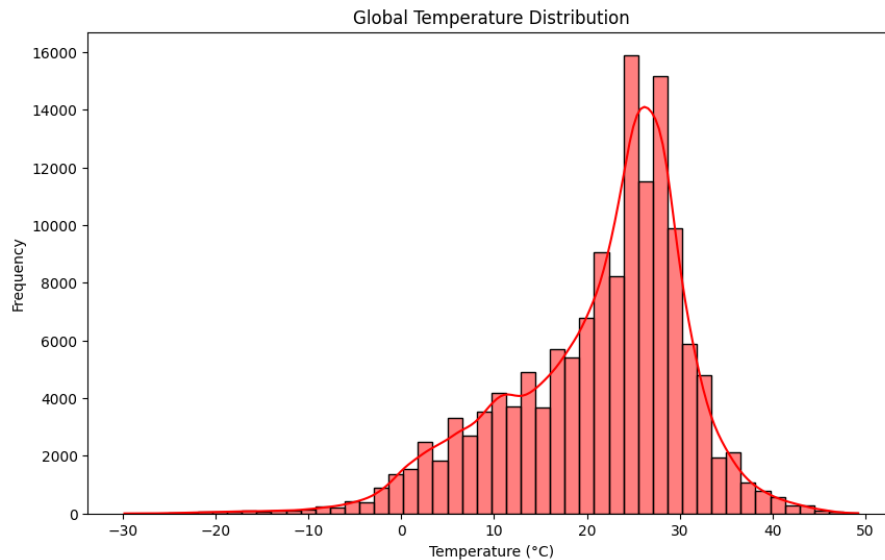


Figure 15: Global Temperature Distribution

Overall, the climate analysis highlights both temporal weather cycles and regional long-term temperature differences across countries.

5.2 Environmental Impact Analysis

To study the relationship between environmental pollution and weather conditions, air quality indicators were analyzed alongside major atmospheric variables. PM2.5 concentration was used as the primary air quality metric due to its significance in public health and environmental monitoring. Variables examined included temperature, humidity, wind speed, pressure, PM2.5, PM10, Carbon Monoxide, Ozone and Nitrogen Dioxide.

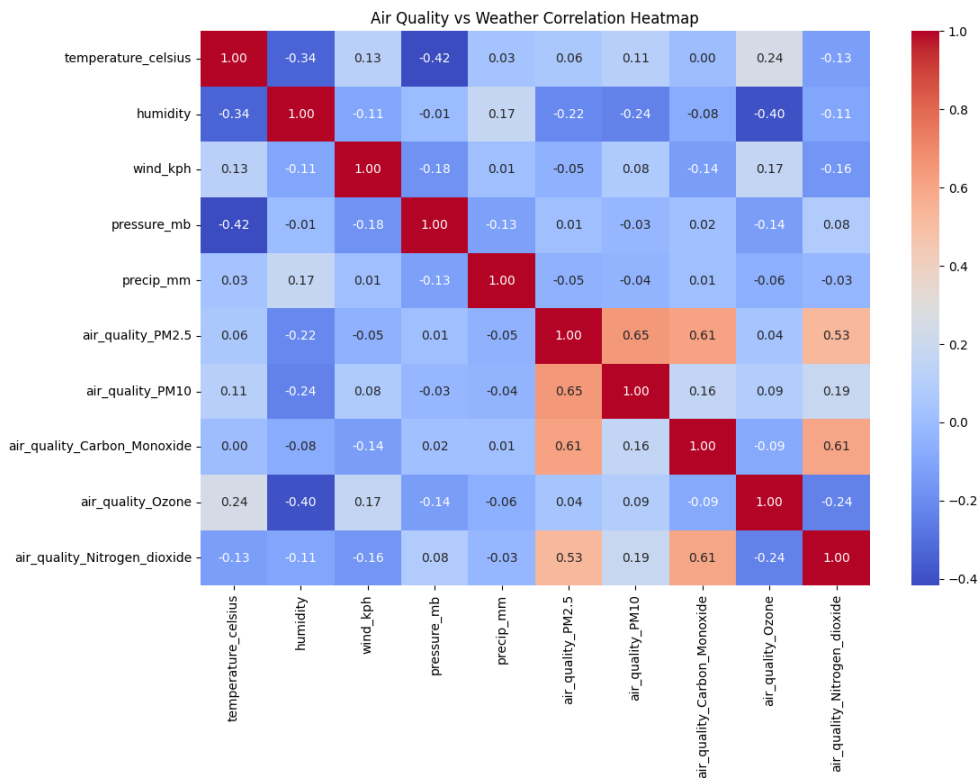


Figure 16: Air Quality vs Weather Correlation Heatmap

The correlation heatmap reveals that PM2.5 and PM10 exhibit weak to moderate correlations with meteorological variables. PM2.5 showed a slight negative relationship with wind speed, suggesting that stronger winds contribute to the dispersion of airborne particulate matter and reduce local pollution concentration. Atmospheric pressure and humidity demonstrated weaker associations with pollution levels.

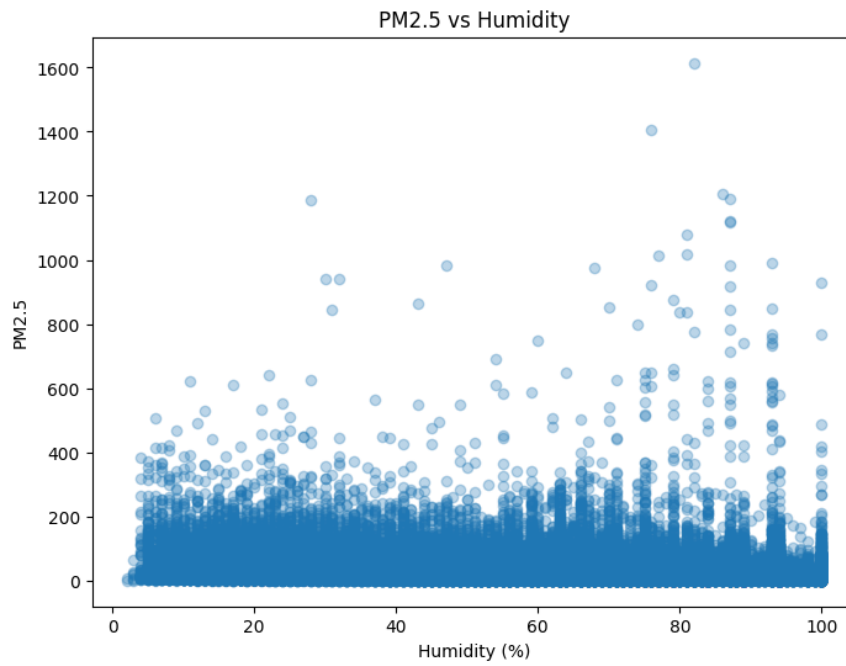


Figure 17: PM2.5 vs Humidity

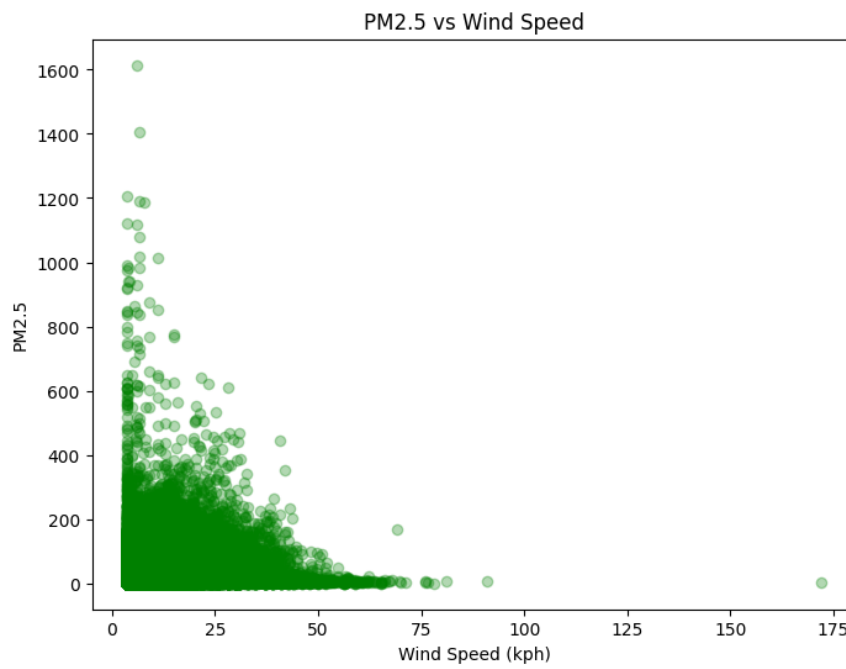


Figure 18: PM2.5 vs Wind Speed

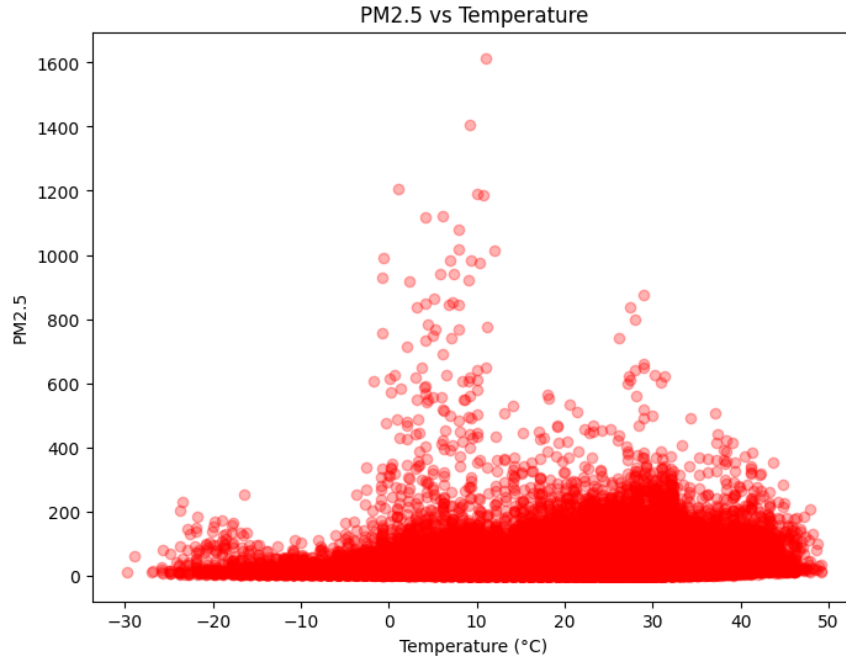


Figure 19: PM2.5 vs Temperature

Overall, this analysis demonstrates that weather conditions can meaningfully influence air quality behaviour. Higher wind speeds generally reduce pollution concentration through dispersion, while higher humidity may be associated with increased particulate accumulation. These relationships highlight the interaction between atmospheric conditions and pollutant behaviour, though the effect magnitudes vary across geographic regions.

5.3 Feature Importance Analysis

To better understand the drivers behind forecasting performance, feature importance analysis was conducted using three complementary methods: tree-based importance from Gradient Boosting, permutation importance and correlation-based analysis.

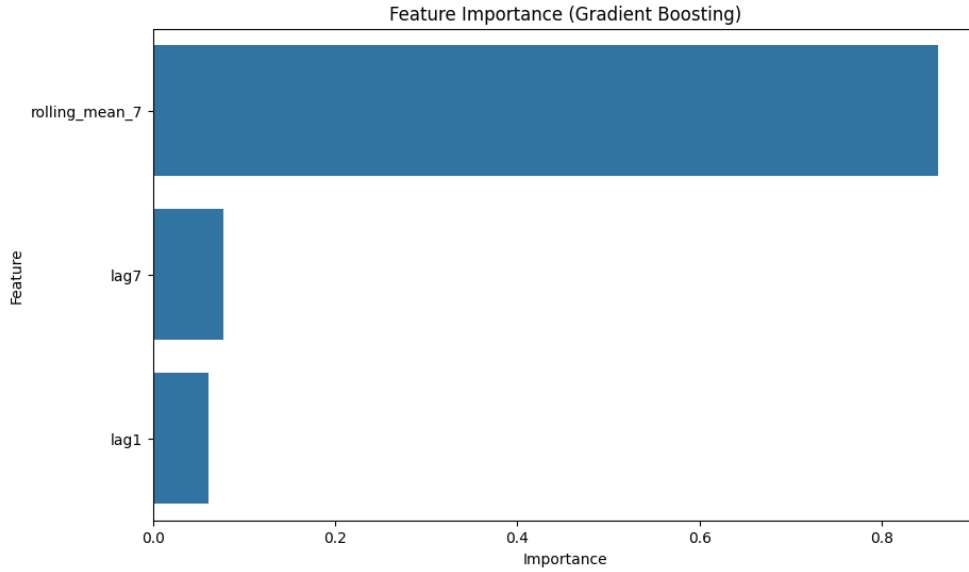


Figure 20: Feature Importance Gradient Boosting (Tree-Based)

The Gradient Boosting feature importance plot identifies the features used most frequently in decision splits across all trees. Lag-based temporal features (lag-1 and 7-day rolling mean temperature) were consistently ranked as the most influential predictors, followed by atmospheric pressure and humidity.

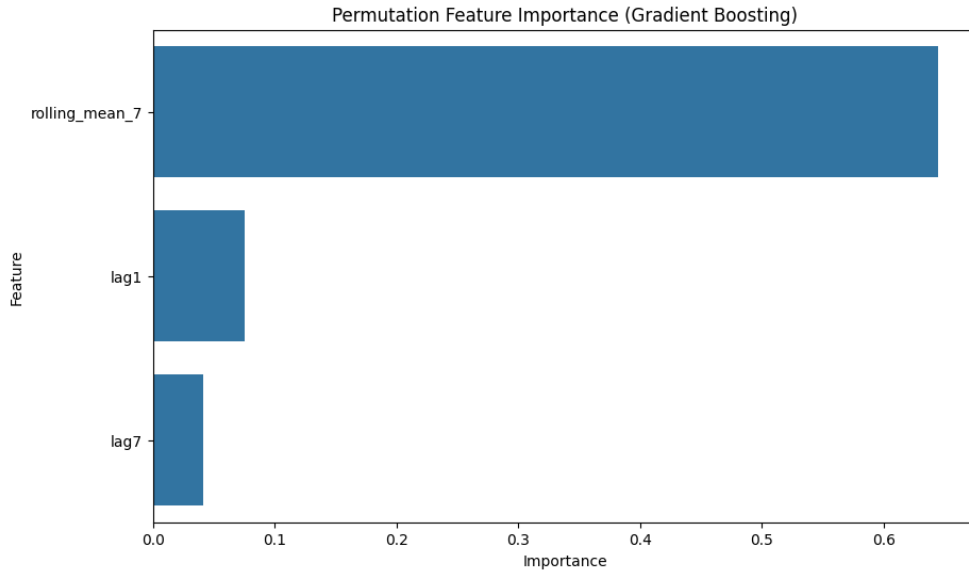


Figure 21: Permutation Feature Importance Gradient Boosting

Permutation importance, which measures the degradation in model performance when each feature is randomly shuffled, produced consistent results. Lag-based temporal features and atmospheric pressure were again the strongest contributors to predictive accuracy. The agreement between both methods increases confidence in the reliability of the model interpretation.

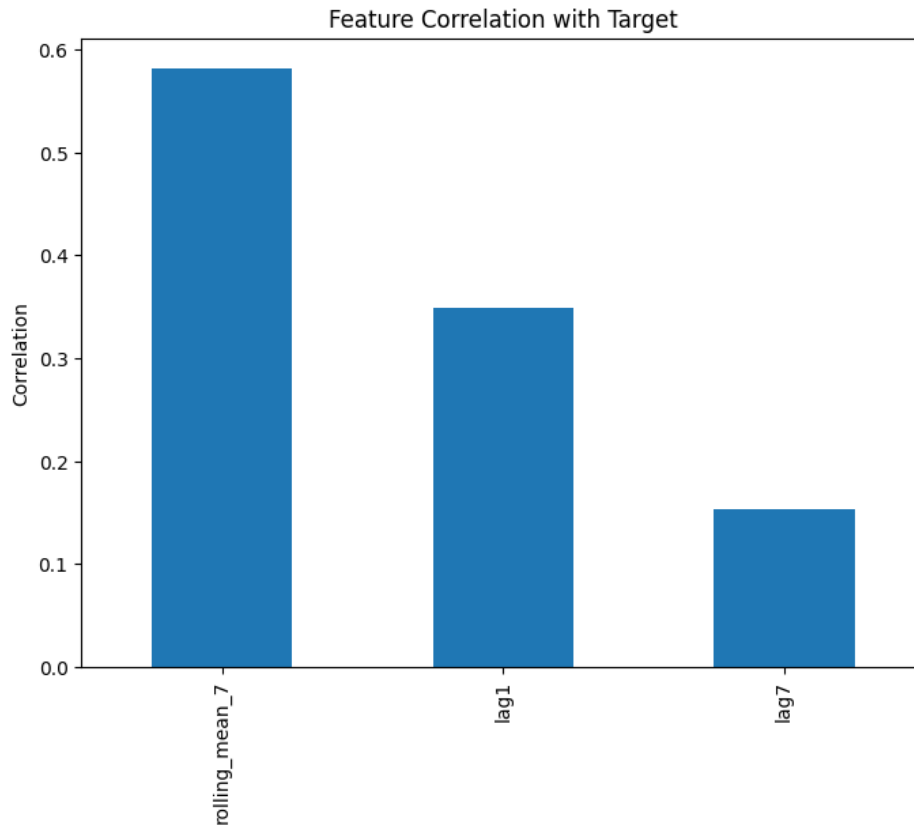


Figure 22: Feature Correlation with Target Temperature

Correlation-based analysis provided additional insight into linear relationships between predictors and the target variable. Across all three methods, the conclusion is consistent: recent temperature history and atmospheric pressure are the most important variables driving model predictions.

5.4 Spatial Analysis

Geographical coordinates were used to visualize how weather conditions vary across different global regions. Spatial analysis helps reveal regional climate behaviour and demonstrates the worldwide coverage of the dataset.

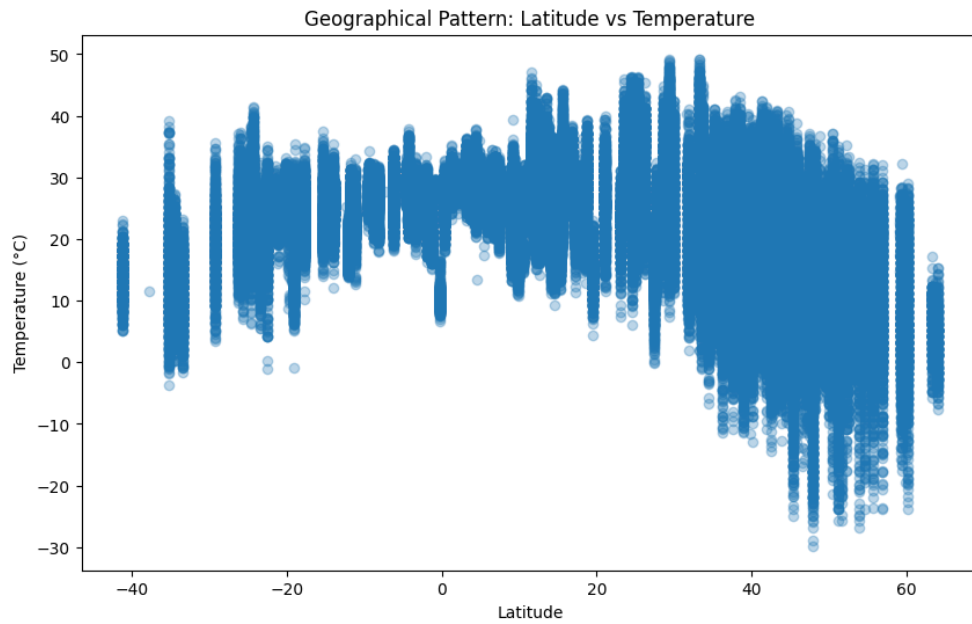


Figure 23: Latitude vs Temperature

Temperature shows a clear and expected dependency on latitude. Regions closer to the equator consistently exhibit higher temperatures, while higher latitudes towards the poles are associated with cooler conditions. This confirms established global climate zoning patterns.

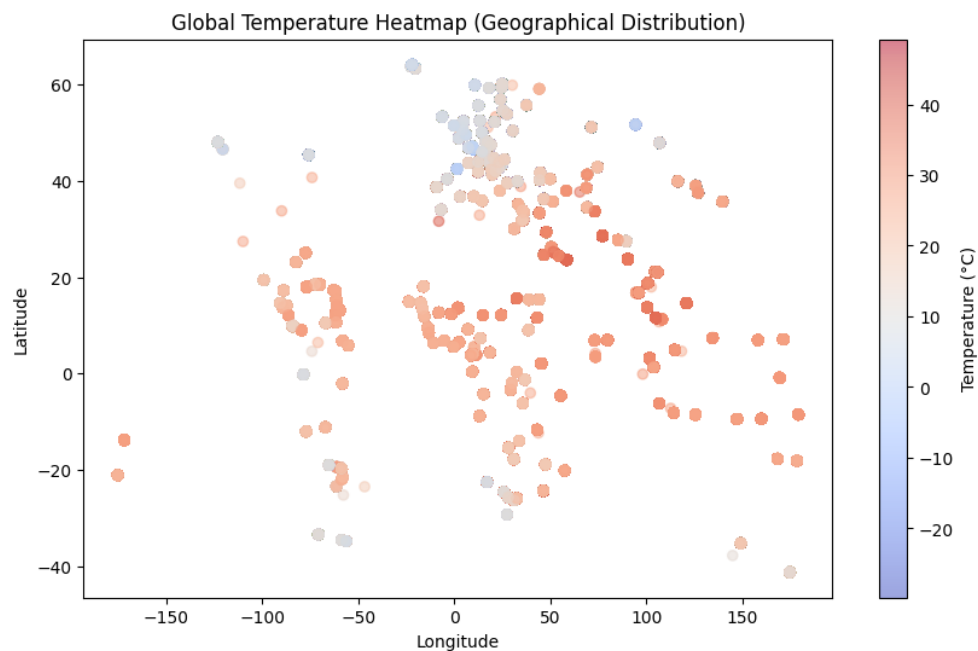


Figure 24: Global Temperature Heatmap (Longitude vs Latitude)

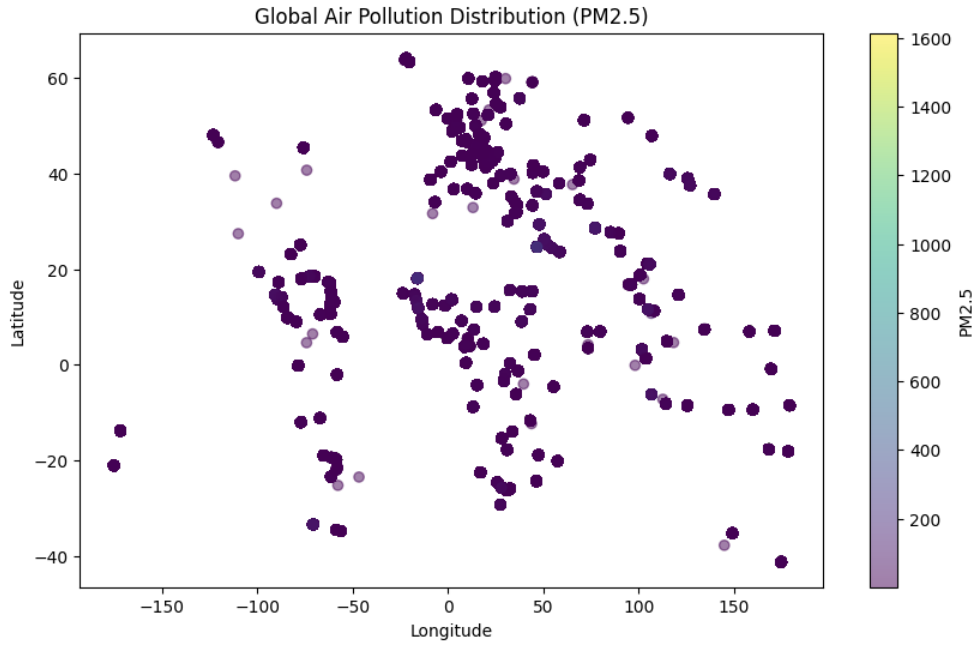


Figure 25: Global Air Pollution Distribution PM2.5 (Longitude vs Latitude)

The global PM2.5 map shows that air pollution levels vary significantly across geographic regions. Densely populated and industrialised areas exhibit notably higher particulate concentrations. The longitude versus temperature distribution also confirms that atmospheric conditions differ substantially across eastern and western hemispheres.

Overall, the spatial analysis confirms that both temperature and air quality are strongly influenced by geography, making geographical coordinates and regional features important inputs for weather forecasting and environmental models.

5.5 Geographical Patterns

To further understand regional climate differences, country-level comparisons were aggregated to the continental level using a custom country-to-continent mapping. Average temperature, precipitation and PM2.5 concentration were computed for each continent represented in the dataset.

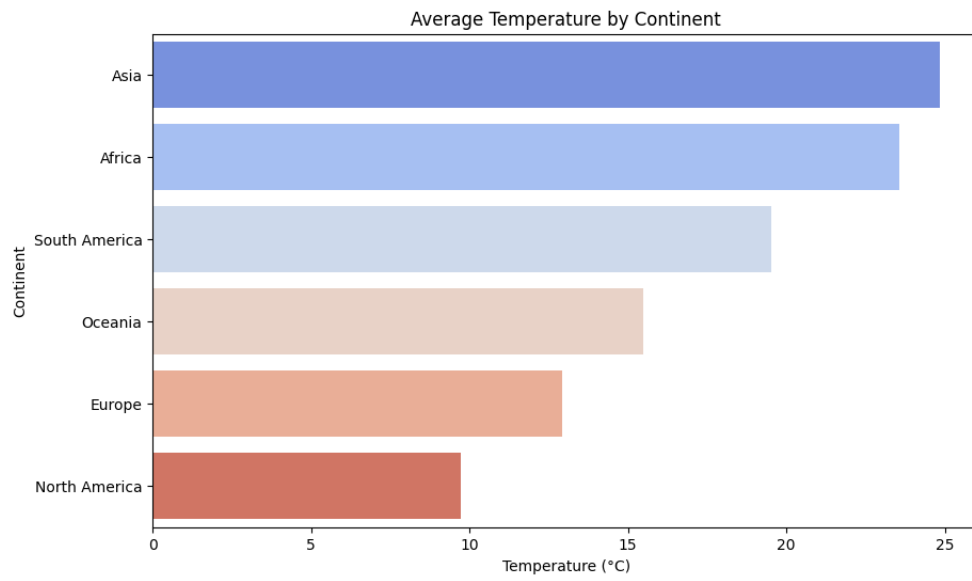


Figure 26: Average Temperature by Continent

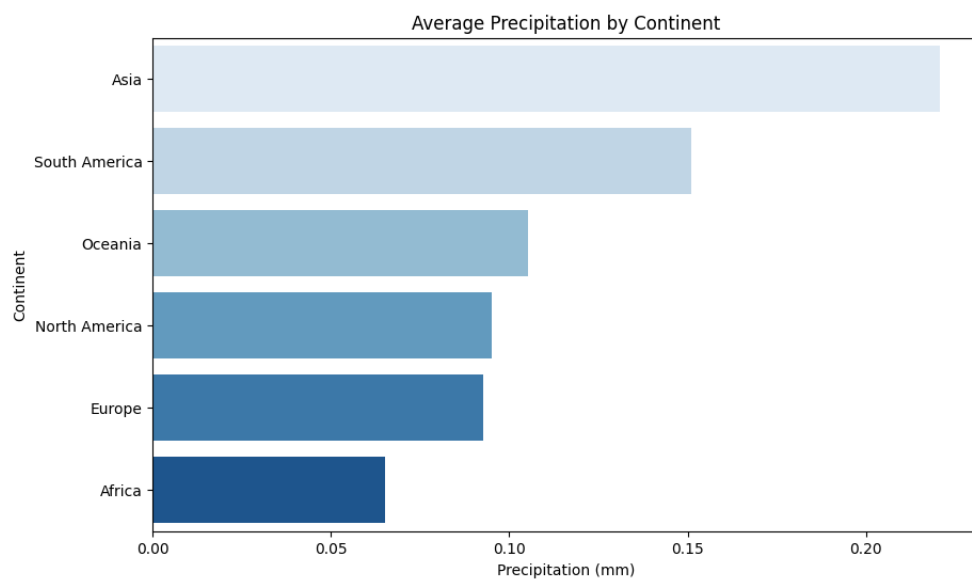


Figure 27: Average Precipitation by Continent

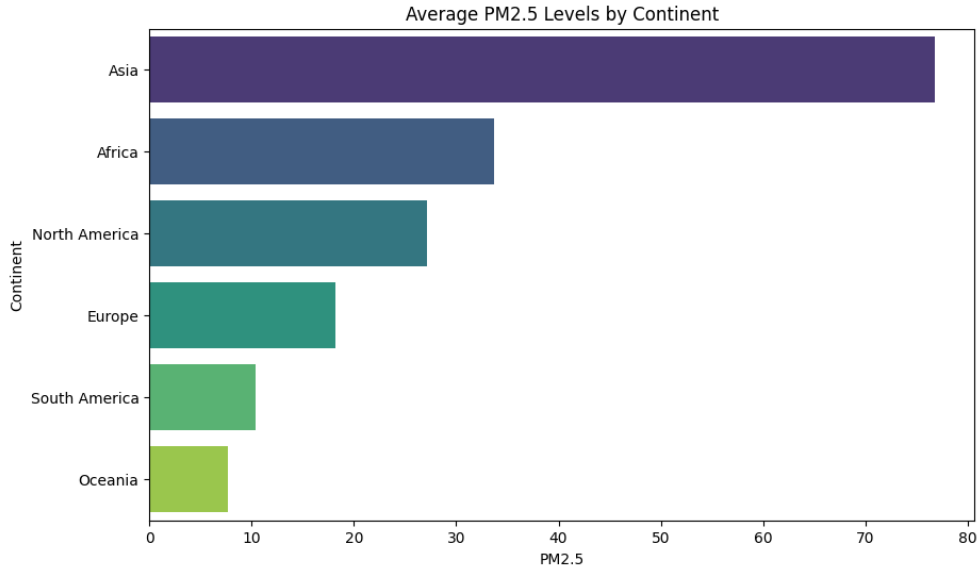


Figure 28: Average PM2.5 Levels by Continent

Tropical and equatorial continents such as Africa, South America and parts of Asia recorded the highest average temperatures, while Europe and North America exhibited more moderate climates. Precipitation patterns varied widely, with some continents experiencing consistently higher rainfall. Air quality also differed markedly across regions, with industrialised and densely populated continents showing higher PM2.5 concentrations. These patterns highlight that climate behaviour is fundamentally tied to geography, and that continental aggregation provides a useful macro-level perspective on global weather variation.

6 Conclusion

This project successfully completed an end-to-end weather analytics and forecasting pipeline using the Global Weather Repository dataset. The workflow encompassed data cleaning, exploratory data analysis, feature engineering, baseline time series forecasting, advanced machine learning model comparison, ensemble learning and five domain-specific analytical studies covering climate trends, environmental impact, feature importance, spatial distribution and geographical patterns.

The forecasting results demonstrated that tree-based ensemble models significantly outperform traditional statistical approaches. Gradient Boosting achieved the strongest individual performance, while the ensemble of Random Forest, Gradient Boosting and XGBoost further improved robustness. ARIMA and Prophet were found to be poorly suited to globally aggregated, multi-region weather data due to violations of their underlying modelling assumptions.

Anomaly detection using Z-Score, IQR and Isolation Forest revealed that multivariate methods are better equipped to identify unusual weather patterns in complex datasets. Air quality analysis showed that meteorological conditions — particularly wind speed — meaningfully influence pollutant concentrations. Feature importance analysis consistently identified recent temperature history (lag features) and atmospheric pressure as the strongest predictors of future temperature. Spatial and continental analyses confirmed that geography is a fundamental driver of both temperature and air quality variation worldwide.

Overall, this assessment demonstrates strong practical skills in data science, machine learning, feature engineering, model evaluation and business-oriented analytical thinking. Future work could include deep learning time series models (e.g. LSTM, Temporal Fusion Transformer), live weather API integration and deployment as an interactive forecasting dashboard.